

Question Paper

Exam Date & Time: 01-Jan-2026 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - JANUARY 2026
REINFORCEMENT LEARNING [CSE_4028]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Model how the key elements of Reinforcement Learning work together in enabling an autonomous robot to navigate a maze. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Application"
- B) Develop the application of Positive and Negative Reinforcement in designing a game-playing agent that learns optimal moves through feedback. (3)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Identify the limitations and scope of the Reinforcement Learning problem when applied to real-world decision-making systems. (2)
- Course Outcome: "CO1" / Cognitive Level: "Application"
- 2) Analyze how the principle of inclusion-exclusion can be applied to find the probability of the union of three events A, B , and C . Derive an expression for $P(A \cup B \cup C)$ in terms of their individual and joint probabilities. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Let the probability density function (PDF) of a continuous random variable X be defined as
- $$f(x) = \begin{cases} kx(2-x), & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$
- Solve for k such that $f(x)$ is a valid probability density function and the cumulative distribution function (CDF) $F(x)$ of X . (3)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- C) Identify how the probability mass function (PMF) is used to describe the behavior of a discrete random variable and determine the probability of specific outcomes. (2)
- Course Outcome: "CO2" / Cognitive Level: "Application"
- 3) Analyze how the Action-Value Methods are applied to solve the n-Armed Bandit Problem, highlighting how different approaches such as the sample-average and incremental methods estimate the action values and influence the exploration-exploitation trade-off. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Apply how the unified notation is applied to represent both episodic and continuing tasks within a finite Markov Decision Process, emphasizing the role of the discount factor in connecting the two formulations. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Apply the concept of Upper-Confidence-Bound (UCB) Variables to select actions in a reinforcement learning task, balancing exploration and exploitation during decision-making. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Apply the concepts of optimality and approximation can be applied in a robotic path-planning scenario, where the agent must learn near-optimal policies under limited computational resources and incomplete environment knowledge. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Compare the processes of Policy Iteration and Value Iteration in Dynamic Programming, emphasizing their computational steps, convergence characteristics, and effectiveness in obtaining the optimal policy. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Develop an example of a finite MDP where the agent optimizes long-term returns to achieve a specific goal state. (2)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) Compare On-Policy and Off-Policy Monte Carlo Control methods, emphasizing how importance sampling enables the off-policy approach to learn an optimal policy while following a different behavior policy. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Apply how Actor-Critic Methods can be modified using eligibility traces to enhance learning efficiency, enabling faster policy and value function updates in dynamic reinforcement learning environment (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Develop a short expression for the R-Learning value update, highlighting how it differs from standard Q-Learning. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4028:REINFORCEMENT LEARNING

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10

5 5 CO 5

10
Total 50

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	CSE_4028:REINFORCEMENT LEARNING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	10	30(60%)
4	Analysis	5	20(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	15	50 (100%)